# TURBULENCE THROUGH CANONICAL FLOWS

Transition, spectra, examples, data, and computation

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Fluid Dynamics and IBM Theory • Module 5

Adapted from a lecture by Steven L. Brunton.

**Guiding idea**

Canonical flows isolate mechanisms that recur in more complicated systems.

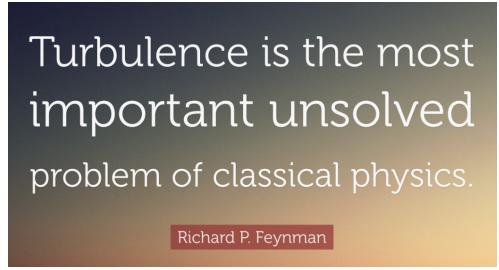
Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

# LECTURE ROADMAP

- Why turbulence is hard (and why it matters)
- Canonical flows: a “vocabulary” for turbulence
- Examples: pipe, wake, boundary layer, cavity, jets, mixing layers
- “More complex” flows (geometry, separation, multi-scale physics)
- Open-source resources: databases and CFD codes

# WHY TURBULENCE IS HARD

- Turbulence is a *multiscale*, nonlinear dynamics problem whose observed state can depend on disturbance history.
- Small perturbations can grow, interact, and reorganize the flow.
- Many flows are *linearly stable* yet become turbulent through finite-amplitude disturbances.
- Predicting transition and statistics (drag, mixing, heat transfer, intermittency) is still challenging.




Quotation graphic (stock “quote image”; originating website to be confirmed). The wording is a popular paraphrase attributed to R. P. Feynman; his own text says turbulence is an old, unsolved problem common to many fields (*Feynman Lectures on Physics*, Vol. I, Ch. 3).

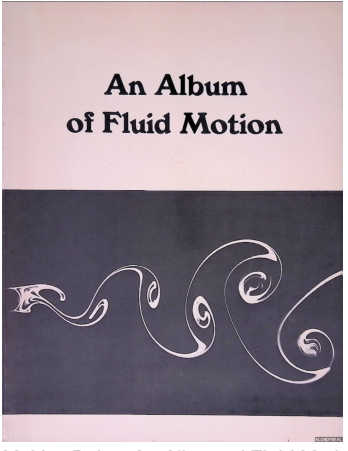
The *Feynman Lectures* (Vol. II, Ch. 41) give a classic discussion of why real viscous flows are difficult to predict.

# CANONICAL FLOWS (VAN DYKE “ALBUM”)

**Idea**

A small set of “canonical” configurations show up *everywhere*. Learn these and you gain a language for:

- transition mechanisms (instability $\rightarrow$ breakdown),
- coherent structures (vortices, streaks, shear layers),
- scaling laws and nondimensional groups.




M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982), cover.

# A MINIMAL TOOL: REYNOLDS NUMBER

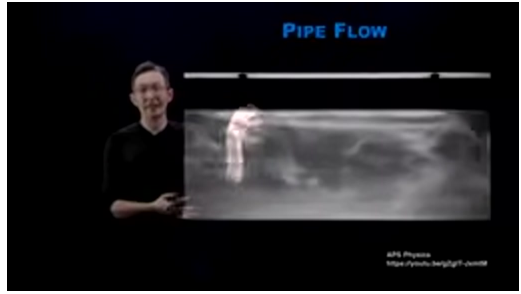
- For chosen velocity and length scales, Reynolds number compares inertia to viscosity:

$$\mathrm{Re} = \frac{UL}{\nu}.$$

- "High" $\mathrm{Re}$ often permits thinner shear layers and wider scale separation, but does not guarantee turbulence.
- Treat $\mathrm{Re}$ as a control parameter, not a universal one-to-one label for complexity: geometry, forcing, roughness, and disturbance history matter.

# PIPE FLOW

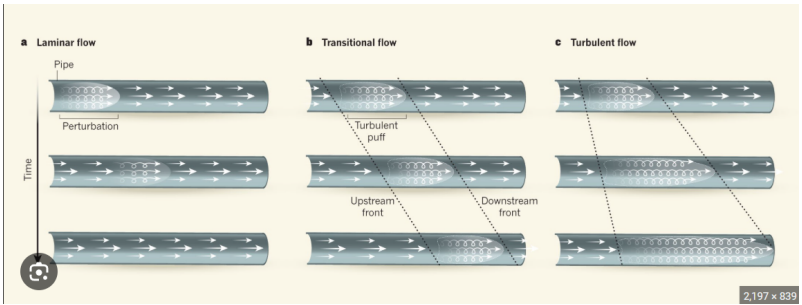
- Classic internal flow: long straight pipe, pressure-driven.
- The Hagen–Poiseuille solution exists for every $\mathrm{Re}$ and is linearly stable (proved for axisymmetric disturbances; computed stable to $\mathrm{Re} = 10^7$, Meseguer & Trefethen 2003), yet finite-amplitude disturbances can trigger turbulence. (Their $\mathrm{Re}$ uses centerline speed and radius, which for the parabolic profile equals $U_b D/\nu$.)
- Transition often involves localized “puffs” and “slugs” (intermittency).
- Active control is possible: carefully designed perturbations can suppress turbulence (drag reduction).




Still from S. L. Brunton’s canonical-flows lecture (linked on the title slide); the embedded pipe-flow dye visualization is credited on screen to an APS Physics video on YouTube (full URL to be confirmed).

# PIPE FLOW: SLUGS AND PUFFS

- Transitional pipe flow can coexist with laminar regions and is intrinsically intermittent.
- Localized turbulent structures have distinct dynamics:
  - **puffs** remain localized while advecting and may decay or split stochastically;
  - at higher $\mathrm{Re}$, **slugs** expand by turbulent invasion at their fronts.
- The crossover is a range of behavior rather than a single universal threshold.



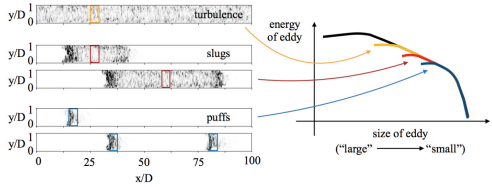

Source: Graham, M. Turbulence spreads like wildfire. Nature 526, 508–509 (2015).
M. D. Graham's News & Views commentary (doi:10.1038/526508a) on D. Barkley et al., "The rise of fully turbulent flow," *Nature* **526**, 550–553 (2015).

# PUFFS, SLUGS, AND SPECTRAL SIMILARITY

- The image sketches spectra sampled from the marked pipe-flow regions.
- For high-$\mathrm{Re}$, locally isotropic **3D** turbulence, Kolmogorov similarity hypothesizes an inertial range

$$E(k) = C_K \varepsilon^{2/3} k^{-5/3}.$$

- This is asymptotic and statistical—not exact universality. Intermittency, anisotropy, walls, forcing, and finite $\mathrm{Re}$ produce deviations.
- A 2D inverse-energy cascade may also have a $-5/3$ range, but its transfer direction and mechanism differ; enstrophy transfers downscale.




Source image: Phys.org (2020).

The news item reports work of the Chakraborty group (OIST) on spectra of puffs and slugs; see also R. T. Cerbus, C.-C. Liu, G. Gioia & P. Chakraborty, *Phys. Rev. Lett.* **120**, 054502 (2018). Original figure credit as given in the news item (to be confirmed).

# WAKE FLOWS

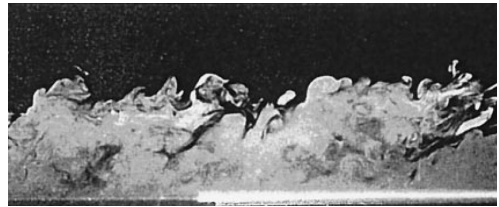
- Flow past a body leaves a *wake* with a velocity deficit and strong shear.
- Bluff bodies can shed a Kármán vortex street, producing unsteady loading and noise; lifting surfaces also generate trailing and tip vortices.
- Wakes are central to:
  - aircraft wing tips and propellers,
  - wind farms and arrays,
  - animal locomotion and swimming.




Still from S. L. Brunton's canonical-flows lecture (linked on the title slide); embedded wake images credited on screen to B. Herrmann, P. Oswald, R. Semaan & S. L. Brunton, *Commun. Phys.* (2020) ("Modeling synchronization in forced turbulent oscillator flows"; open access, CC BY 4.0; details to be confirmed).

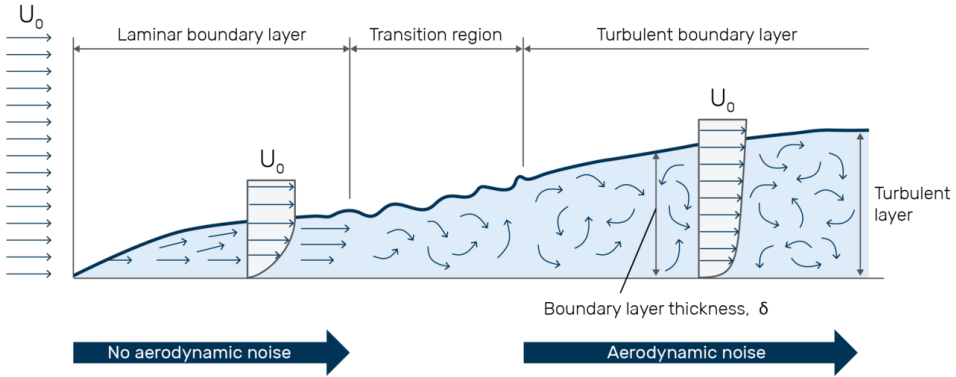
# BOUNDARY LAYER

- Near a wall, no-slip forces a thin region of strong shear: the boundary layer.
- Key questions:
  - when does laminar flow transition to turbulence?
  - how do near-wall streaks and vortices sustain turbulence?
  - how do roughness and pressure gradients change drag?
- Boundary layers dominate drag and heat transfer on vehicles, buildings, and animals.



Smoke visualization of a turbulent boundary layer; Van Dyke, *An Album of Fluid Motion* (turbulent boundary layer plate; also reproduced in Dou 2018).
Photographer (likely R. E. Falco) and plate number to be confirmed. Note the sharp, convoluted edge between turbulent (smoke-filled) and irrotational outer fluid: boundary-layer intermittency.

# BOUNDARY LAYER: SPATIAL DEVELOPMENT

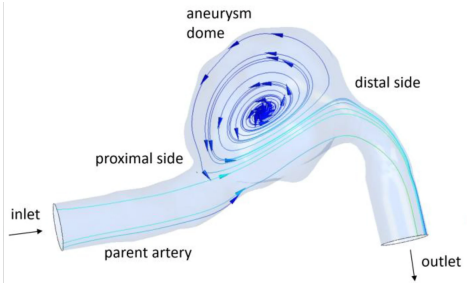



Boundary-layer schematic after GRAS Sound & Vibration, *The boundary layer—laminar and turbulent flows* (blog; source to confirm), https://www.grasacoustics.com/blog/aerodynamic-measurements-components-of-the-boundary-layer.

Reading the figure: the boundary-layer thickness $\delta$ grows as $\delta \propto x/\mathrm{Re}_x^{1/2}$ in the laminar region (Blasius) and faster, roughly $\delta \propto x/\mathrm{Re}_x^{1/5}$, once turbulent; on a smooth flat plate transition typically occurs for $\mathrm{Re}_x = U_0 x/\nu \sim 3\times10^5$–$3\times10^6$, depending on free-stream disturbances. The “no aerodynamic noise” label is a simplification: laminar layers radiate far less, not nothing.

# CAVITY FLOW

- In an **open engineering cavity**, a shear layer spans the opening; pressure/acoustic feedback can create resonant oscillations.
- An intracranial aneurysm is instead a **cavity-like recirculating flow**: pulsatility, geometry, and vessel-wall conditions shape its hemodynamics.
- Related applications:
  - aneurysm inflow jets and recirculation,
  - gaps, steps, and surface defects,
  - flow-induced vibration.




Source: Boniforti MA, Magini R, Orosco Salinas T. Hemodynamic Investigation of the Flow Diverter Treatment of Intracranial Aneurysm. Fluids. 2023; 8(7):189.

# JETS, MIXING LAYERS, AND SEPARATED FLOWS (OVERVIEW)

- **Round jet:** downstream of the potential core the mean profile becomes *self-similar*, the half-width grows roughly linearly, and Kelvin–Helmholtz roll-up of the annular shear layer seeds the initial vortex rings. The half-velocity radius grows as $\mathrm{d}r_{1/2}/\mathrm{d}x \approx 0.09$–$0.10$ (about $5$–$6^\circ$; Pope 2000, §5.1); the visible turbulent edge spreads at a half-angle of order $10$–$12^\circ$.
- **Plane mixing layer:** two streams of different speed produce the Brown–Roshko coherent spanwise rollers, which pair and merge; at high $\mathrm{Re}$ the thickness grows linearly, $\mathrm{d}\delta/\mathrm{d}x \propto \lambda = (U_1 - U_2)/(U_1 + U_2)$, and is essentially independent of $\mathrm{Re}$ (and only weakly dependent on the density ratio).

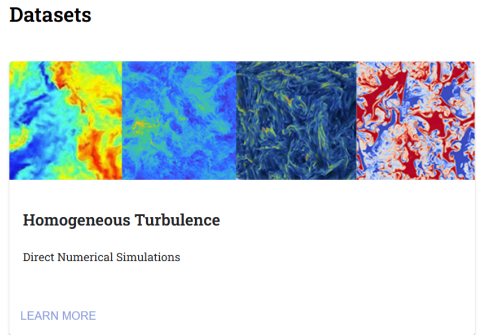
> **Self-similar** means one profile shape after rescaling: for the round jet $\bar{U}(r,x)/U_0(x) = f\big(r/r_{1/2}(x)\big)$, with centreline velocity $U_0(x) \propto 1/x$ and $r_{1/2} \propto x$, so the momentum flux $\propto U_0^2 r_{1/2}^2$ stays constant (Pope 2000, §5.1).

# OVERVIEW, CONTINUED: SEPARATED AND COMBINED FLOWS

- **Separated flows:** a backward-facing step separates at a fixed geometric edge; at high $\mathrm{Re}$ the mean reattachment length is about $6$–$7$ step heights ($7 \pm 1$ in the review of Eaton & Johnston 1981, varying with expansion ratio and the incoming boundary layer; laminar steps reattach at an $\mathrm{Re}$-dependent distance), and the shear layer, recirculation bubble, and reattachment zone each have their own statistics.
- **Combinations:** real configurations superpose these—a jet issuing into a cavity, a wake feeding a mixing layer, a step downstream of a boundary layer.
- **Multi-scale physics:** geometry sets the large scales; viscosity sets the small ones. The gap between them is what makes prediction expensive.

# OPEN DATA: TURBULENCE DATABASES

- Public datasets let you test ideas without running a massive DNS yourself.
- Example: Johns Hopkins Turbulence Databases (JHTDB)
  - isotropic turbulence, channel flow, boundary layers, geophysical, . . .
  - query-able via web/API-style interfaces
  - great for teaching statistics, structure functions, spectra, and coherent-structure detection



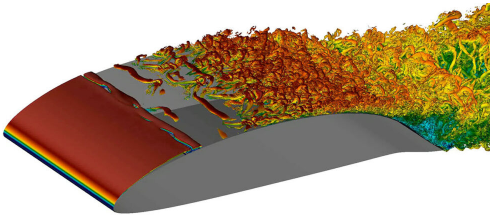

Johns Hopkins Turbulence Databases, turbulence.pha.jhu.edu.

# OPEN-SOURCE CFD CODES

**Two common ecosystems**

- **OpenFOAM**: open-source finite-volume framework with many RANS/LES solvers.
- **Nek5000**: high-order spectral-element code; widely used for DNS/LES in complex geometries.

- Choosing a tool depends on: accuracy needs, geometry, turbulence model, and HPC resources.



Nek5000 project banner image, Argonne National Laboratory, nek5000.mcs.anl.gov.

# TAKEAWAYS

- Turbulence is everywhere, but canonical flows give structure to the chaos.
- Each canonical flow highlights different mechanisms (instability, shear, walls, separation, resonance).
- Open videos, datasets, and codes make it easier than ever to *see* turbulence and to explore it quantitatively.

# SELECTED REFERENCES AND SOURCE NOTES


- R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. II, Ch. 41, “The Flow of Wet Water”.
- K. Avila, D. Moxey, A. de Lozar, M. Avila, D. Barkley, and B. Hof, “The onset of turbulence in pipe flow,” *Science* **333** (2011), 192–196. doi:10.1126/science.1203223.
- A. N. Kolmogorov, “The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers,” *Proc. R. Soc. A* **434** (1991 translation), 9–13. doi:10.1098/rspa.1991.0075.
- R. H. Kraichnan, “Inertial ranges in two-dimensional turbulence,” *Phys. Fluids* **10** (1967), 1417–1423. doi:10.1063/1.1762301.
- H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed., Springer (2017).
- Official resources: JHTDB, OpenFOAM Foundation, and Nek5000.

# SELECTED REFERENCES AND SOURCE NOTES (CONTINUED)


- S. L. Brunton, lecture on canonical turbulent flows (source of the slide sequence and of the two lecture stills), Cassyni seminar page.
- M. Van Dyke, *An Album of Fluid Motion*, Parabolic Press (1982).
- S. B. Pope, *Turbulent Flows*, Cambridge University Press (2000), Ch. 5 (jets and mixing layers) and Ch. 7 (wall flows).
- A. Meseguer and L. N. Trefethen, "Linearized pipe flow to Reynolds number $10^7$," *J. Comput. Phys.* **186** (2003), 178–197. doi:10.1016/S0021-9991(03)00029-9.
- D. Barkley, B. Song, V. Mukund, G. Lemoult, M. Avila, and B. Hof, "The rise of fully turbulent flow," *Nature* **526** (2015), 550–553. doi:10.1038/nature15701; commentary: M. D. Graham, *Nature* **526** (2015), 508–509.
- G. L. Brown and A. Roshko, "On density effects and large structure in turbulent mixing layers," *J. Fluid Mech.* **64** (1974), 775–816. doi:10.1017/S002211207400190X.
- J. K. Eaton and J. P. Johnston, "A review of research on subsonic turbulent flow reattachment," *AIAA J.* **19** (1981), 1093–1100. doi:10.2514/3.60048.
- B. Herrmann, P. Oswald, R. Semaan, and S. L. Brunton, "Modeling synchronization in forced turbulent oscillator flows," *Commun. Phys.* **3** (2020), 195 (wake images in the lecture still).
- R. T. Cerbus, C.-C. Liu, G. Gioia, and P. Chakraborty, "Kolmogorovian turbulence in transitional pipe flows," *Phys. Rev. Lett.* **120** (2018), 054502.
- M. A. Boniforti, R. Magini, and T. Orosco Salinas, "Hemodynamic investigation of the flow diverter treatment of intracranial aneurysm," *Fluids* **8** (2023), 189 (open access).
- Remaining image credits (Phys.org, GRAS, JHTDB, Nek5000) are on the slides where they appear.